ENGINEERING
PROJECT PORTFOLIO

Healthcare AI, data platforms, cloud modernization, digital innovation, and enterprise software delivery

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PORTFOLIO BRIEF

Representative work across AI-enabled healthcare platforms, cloud data systems, enterprise modernization, workforce automation, and engineering leadership.

12+ featured case studies
50+ IT programs
100+ hands-on projects

PORTFOLIO THEMES

The through-lines that connect the projects

AI AND AGENTIC SYSTEMS
Retrieval, validation, evaluation loops, human review, usage controls, and practical AI delivery.

HEALTHCARE AND REGULATED PLATFORMS
HIPAA-regulated SaaS, EHR workflows, interoperability, clinical analytics, and support readiness.

CLOUD AND DATA MODERNIZATION
Azure, AWS, Databricks, data lakes, search, CI/CD, serverless patterns, and microservices.

ENGINEERING MANAGEMENT
Distributed teams, solution design, estimation, Agile execution, mentoring, and transition to operations.

DIGITAL INNOVATION
Discovery workshops, prototypes, MVP planning, automation, workplace modernization, and measurable value.

PROJECT INDEX

A quick map of the portfolio

PROJECT	SECTION	SIGNAL	ROLE
Novare | AI-powered ambulatory care platform	Featured	Supported platform modernization and AI-enabled workflow delivery across EHR, practice management, revenue-cycle, ambien...	Served as Principal Engineer and technical lead across Clinical AI, Shared Data Platform, Data Insights, and Analytics in a HIPAA-regulated healthcare SaaS environment.
Greenway Practice Analytics and Reporting Platform | Healthcare analytics, revenue intelligence, and value-based care reporting	Featured	Improved visibility into productivity, revenue, patient trends, care gaps, value-based program progress, MIPS and UDS reporting, and near-real-time A/...	Served as Principal Engineer and technical lead for a Greenway analytics and reporting platform supporting practice performance, revenue-cycle visibility, and value-based care insights.
FHIR Health Equity Pipeline | Local-first healthcare data engineering and analytics demo	Public Demo	Demonstrated 347 synthetic FHIR resources across 60 patients, 39 dbt tests, care-gap marts, pipeline monitor workflows, FHIR R4/R4B/R5 mapping decisio...	Served as Principal Engineer, technical lead, architect, and hands-on builder for a public portfolio-grade healthcare data platform demo.

PROJECT	SECTION	SIGNAL	ROLE
Application Performance Monitoring with .NET and OpenTelemetry | Reusable observability enablement for .NET applications	Observability	Reduced observability adoption friction, gave teams a repeatable integration path, improved troubleshooting readiness, and helped surface performance,...	Served as Principal Engineer, technical lead, solution architect, and hands-on builder for a reusable .NET observability solution and implementation guide.
Real-Time ECS Cost Optimization and Downscaling Automation | Java-based cloud cost-control automation for TaxAct ECS workloads	Cost Optimization	Helped target tens of thousands of dollars in cloud savings by reducing idle ECS capacity, improving scaling discipline, and preserving workload relia...	Served as Principal Engineer, technical lead, solution architect, and hands-on builder for a Java service that automated real-time ECS capacity reduction.
XPert Booking and Scheduling Platform | Tax expert appointment, calendar, and scheduling platform delivery	Customer Experience	Delivered a stronger expert scheduling capability that improved customer-to-expert appointment workflows, reduced manual coordination, strengthened ca...	Served as Principal Engineer, technical lead, solution architect, and product-technology advisor for the implemented XPert booking and scheduling platform.
Virtual Assistant Recruiting and Employee Automation Platform | AI assistant delivery for recruiter productivity and employee engagement	Digital Innovation	Augmented staff recruiters and employees with an always-available assistant that reduced repetitive work, improved throughput, scaled support interact...	Served as Principal Engineer, technical lead, solution architect, and digital innovation strategist for the implemented virtual assistant platform.
AI Chatbots CHIP and AVA Talent Engagement Platform | Conversational AI delivery for employee support and recruiting	Digital Innovation	Improved chat-based employee support and recruiting automation, made organizational knowledge more accessible, expanded self-service interactions, and...	Served as Principal Engineer, technical lead, solution architect, and innovation strategist for implemented employee-facing chatbot platforms supporting talent engagement.
RunbookOps AI | Reviewable incident-triage workflow	Featured	Demonstrates practical use of retrieval, structured model output, validation, evaluation, human oversight, and maintaina...	Served as Principal Engineer, technical lead, designer, and builder for a public GitHub proof of concept demonstrating hands-on AI engineering capability.
CostOps Gateway and LLM Evaluation | Model execution controls	Featured	Shows the governance and platform thinking needed to move model calls from experiments toward controlled application inf...	Served as Principal Engineer, technical lead, designer, and builder for a public GitHub proof of concept for cost-aware LLM execution controls.
TaxAct XPert Assist, XPert Full-Service, and Data Import Platforms	Featured	Improved platform agility, supported more accurate data import, reduced manual entry, and helped teams modernize custome...	Served as Principal Engineer and technical lead for architecture and delivery across flagship customer-facing tax, expert-assist, and data-import experiences.
AI/ML Well Phase Prediction and Logistics Analytics Platform	Featured	Enabled better planning for rig schedules, job details, logistics demand, freight cost control, supplier operations, and...	Served as Principal Engineer, technical lead, and solution architect for a predictive analytics platform within a large drilling, completions, and logistics program.

PROJECT	SECTION	SIGNAL	ROLE
Demand Connect and Deliver Program | Digitally enabled value-chain and logistics platform	Featured	Supported a digitally enabled value chain intended to unlock more than $1B in value through integration, data quality, analytics-driven insights, auto...	Served as Principal Engineer, technical lead, and solution architect for the Demand Connect product vision, architecture, and delivery roadmap.
Cloud Modernization of Loan Origination Platform	Featured	Supported transition to a modern cloud model, improved delivery coordination across 26 scrum teams, and helped establish...	Served as Principal Engineer, technical lead, solution architect, and senior program lead across a cloud modernization program involving many scrum teams.
Business Intelligence Platform Modernization	Featured	Delivered a more complete end-to-end analytics platform and helped move the client toward a maintainable reporting opera...	Served as Principal Engineer, technical lead, and solution architect from envisioning through solution design and implementation oversight.
eFiles Lockbox Processor and e-Vision OCR	Featured	Automated routing and retention of healthcare billing documents, reduced reliance on paper retention, and improved searc...	Served as Principal Engineer, technical lead, solution architect, and delivery lead for a healthcare billing automation solution.
Express Digital Innovation Strategy and Application Modernization	Innovation	Created or shaped roughly 35 innovation opportunities, including AI-powered recruiting, employee career-path analytics, ...	Served as technical lead, solution architect, digital innovation strategist, and delivery lead.
SMS Time Reporting and Payroll Workflow Modernization	Innovation	Outlined a path from paper forms toward more reliable, scalable, accurate time submission and payroll readiness.	Served as technical lead for the solution strategy and modern timecard concept for reducing paper-based time reporting.
E-Documents Initiative and Master Data Foundation	Innovation	Defined a 12-to-15 week pilot path that could reduce duplicate work, improve data fidelity, centralize audit and control...	Served as technical lead for the pilot and phased delivery model for digital employee forms and associate paperwork.
Enterprise Virtual Assistant Concept	Innovation	Created a phased approach for workshops, feasibility assessment, architectural blueprint, pilot development, bot trainin...	Served as technical lead for the virtual assistant concept and pilot approach for supporting recruiters and employees.
Digital Workplace Insights	Innovation	Presented a model for turning workplace data into targeted change efforts, improved adoption, and better return from dig...	Served as technical lead for the workplace analytics story connecting collaboration adoption, digital behaviors, and measurable business value.
Enterprise Content and Collaboration Modernization	Modernization	Supported migration of 5.5TB of content, modernization of collaboration capabilities, restructuring of enterprise knowle...	Served as Principal Engineer, technical lead, solution architect, and delivery lead.
Medicaid and Medicare Program Platform Modernization	Modernization	Improved online presence, functionality, user experience, and continuity of service for a healthcare program administrat...	Served as Principal Engineer, technical lead, solution architect, digital innovation strategist, and delivery lead.

PROJECT	SECTION	SIGNAL	ROLE
Salesforce Document Management and Workflow Platform	Modernization	Created a document and collaboration platform for 50,000+ site collections, improved document traceability, reduced manu...	Served as Principal Engineer, technical lead, solution architect, and delivery lead.

PROJECT	SECTION	SIGNAL	ROLE
Enterprise Search Modernization	Modernization	Supported crawl indexes exceeding 40 million documents and sites and improved access to business-critical knowledge asse...	Served as Principal Engineer, technical lead, solution architect, and delivery lead.
Enterprise Workflow and Batch Framework	Modernization	Helped streamline manual processes, improve insight into financial data, and support one of the client's most ambitious ...	Served as Principal Engineer, technical lead, solution architect, and delivery lead.
Athena Knowledge Management and Learning System	Modernization	Created an early hands-on AI-oriented delivery story that connects expert knowledge, structured learning data, and appli...	Served as Principal Engineer, technical lead, solution architect, and lead programmer.
VendorView Vendor and Contract Management Platform	Modernization	Improved visibility into vendor and contract data, supported acquisition rationalization, and reduced operational fricti...	Served as Principal Engineer, technical lead, and solution architect.
HMO Cloud Migration and Business Process Modernization | Healthcare operations transformation	Modernization	Helped move mission-critical reporting and operational workflows toward a more scalable, cloud-oriented business process platform.	Served as Principal Engineer, technical lead, solution architect, and delivery lead.
Veritas Internal App Store | Enterprise application discovery platform	Modernization	Improved internal application discoverability, reduced friction in finding relevant tools, and supported knowledge sharing around enterprise applicati...	Served as Principal Engineer, technical lead, solution architect, and delivery lead.
Enterprise CMS and Intranet Portal Modernization | Corporate intranet migration and employee self-service	Modernization	Supported migration of more than 15,000 user accounts and personal sites, improved intranet usability, and modernized collaboration and self-service c...	Served as Principal Engineer, technical lead, solution architect, migration lead, UX designer, and delivery lead.
Self-Service Business Intelligence Enterprise Platform | Internal BI architecture and secure data collaboration	Bi And Knowledge	Helped establish a self-service BI capability that accelerated data flow, improved collaboration, and supported better decision-making across business...	Served as Principal Engineer, technical lead, solution architect, and delivery lead.
Consolidated Business Intelligence and Report Distribution Platform | Cross-functional information sharing portal	Bi And Knowledge	Improved access to business data assets, reduced time spent locating reports, strengthened information governance, and supported faster leadership dec...	Served as Principal Engineer, technical lead, solution architect, and delivery lead.
Travel BI Reporting and Analytics Solution | Travel spend, compliance, and sustainability analytics	Bi And Knowledge	Delivered a scalable analytics solution that helped monitor travel spend, improve policy visibility, and support cost and sustainability management.	Served as Principal Engineer, technical lead, solution architect, and delivery lead.

FEATURED CASE STUDIES

Case studies are Word-native, editable, and ready to refine for GitHub publication

01

FEATURED
Healthcare AI
EHR
Practice management
Revenue cycle
Shared data platform

Novare | AI-powered ambulatory care platform

Greenway Health proprietary platform | greenwayhealth.com/novare

Role: Served as Principal Engineer and technical lead across Clinical AI, Shared Data Platform, Data Insights, and Analytics in a HIPAA-regulated healthcare SaaS environment.

Problem: Greenway needed to unify clinical, financial, administrative, and data workflows while embedding AI into a modern ambulatory care platform.

Approach: Helped align product priorities, architecture, software engineering, data engineering, QA, DevOps, vendors, and distributed delivery teams. Advanced shared data services, roles and permissions, AI-assisted workflows, reusable automation, and supportable delivery practices.

Impact: Supported platform modernization and AI-enabled workflow delivery across EHR, practice management, revenue-cycle, ambient documentation, chart preparation, coding support, eligibility, prior authorization, and care-team experiences.

Technologies: Clinical AI, shared data services, HIPAA-regulated delivery, EHR workflows, FHIR-aware integration patterns, API-first design, healthcare analytics.

02

FEATURED
Healthcare analytics
Revenue cycle
AWS / ECS
Databricks

Greenway Practice Analytics and Reporting Platform

Healthcare analytics, revenue intelligence, and value-based care reporting

Role: Served as Principal Engineer and technical lead for a Greenway analytics and reporting platform supporting practice performance, revenue-cycle visibility, and value-based care insights.

Problem: Practices needed consolidated visibility into financial, operational, and clinical performance without relying on disconnected reports, manual spreadsheets, or delayed revenue-cycle insight.

Approach: Led technical direction for a cloud-hosted analytics platform using AWS, ECS, Docker, S3, and Databricks, aligning data ingestion, consolidation, dashboards, reporting workflows, alerts, quality-program tracking, and revenue analytics across EHR and practice-management data domains.

Impact: Improved visibility into productivity, revenue, patient trends, care gaps, value-based program progress, MIPS and UDS reporting, and near-real-time A/R, denials, clean-claim, charges, payments, and adjustment analytics.

Technologies: AWS, ECS, Docker, S3, Databricks, healthcare analytics, EHR and practice-management data, revenue-cycle analytics, dashboarding, custom reporting, MIPS, UDS, value-based care, automated reporting workflows.

03

PUBLIC DEMO
FHIR
Health equity
dbt / DuckDB
Data quality

FHIR Health Equity Pipeline

Local-first healthcare data engineering and analytics demo

Role: Served as Principal Engineer, technical lead, architect, and hands-on builder for a public portfolio-grade healthcare data platform demo.

Problem: Healthcare analytics teams need trustworthy multi-source clinical data pipelines that can validate FHIR-shaped records, protect sensitive data, expose care-gap signals, and show pipeline reliability without using real patient data.

Approach: Built a local-first synthetic FHIR pipeline that ingests simulated source records, lands raw data into bronze models, quarantines invalid payloads, transforms data with dbt into silver and gold marts, masks patient-level analytics, generates quality triage reports, and surfaces health-equity and reliability views in dashboard and mapping workbench artifacts.

Impact: Demonstrated 347 synthetic FHIR resources across 60 patients, 39 dbt tests, care-gap marts, pipeline monitor workflows, FHIR R4/R4B/R5 mapping decisions, privacy-minimization evidence, last-known-good snapshots, quarantine review paths, and OpenTelemetry-style trace visibility.

Technologies: FHIR, Python, dbt, DuckDB, PowerShell, synthetic healthcare data, data quality gates, bronze/silver/gold modeling, care-gap analytics, health-equity dashboards, privacy masking, lineage, quarantine handling, OpenTelemetry-style tracing.

04

OBSERVABILITY
.NET
OpenTelemetry
AWS CloudWatch
AWS X-Ray

Application Performance Monitoring with .NET and OpenTelemetry

Reusable observability enablement for .NET applications

Role: Served as Principal Engineer, technical lead, solution architect, and hands-on builder for a reusable .NET observability solution and implementation guide.

Problem: Teams needed a consistent way to instrument .NET applications, capture actionable telemetry, and standardize observability practices without heavy code changes or one-off project implementations.

Approach: Designed an OpenTelemetry-based enablement package and guidance site covering default and custom instrumentation, service collection extensions, collectors, exporters, metrics, logs, traces, baggage, AWS Distro integration, CloudWatch metrics, X-Ray tracing, implementation roadmaps, and integration patterns for existing .NET projects.

Impact: Reduced observability adoption friction, gave teams a repeatable integration path, improved troubleshooting readiness, and helped surface performance, resiliency, dependency, and runtime signals through standardized telemetry collection and visualization patterns.

Technologies: .NET, OpenTelemetry, AWS Distro for OpenTelemetry, AWS CloudWatch, AWS X-Ray, OpenTelemetry Collector, collectors, exporters, metrics, logs, traces, baggage, service collection extensions, instrumentation libraries.

05

COST OPTIMIZATION
Java
AWS ECS
Fargate
CloudWatch

Real-Time ECS Cost Optimization and Downscaling Automation

Java-based cloud cost-control automation for TaxAct ECS workloads

Role: Served as Principal Engineer, technical lead, solution architect, and hands-on builder for a Java service that automated real-time ECS capacity reduction.

Problem: TaxAct needed to reduce unnecessary AWS compute spend while protecting active workloads from unsafe termination during scaling events and operational updates.

Approach: Built a Java automation solution using AWS ECS, Fargate, SQS, CloudWatch, RDS/SQL, and ECS task-protection patterns to monitor workload signals, coordinate queue-driven processing, and downscale ECS services in real time when capacity was no longer required.

Impact: Helped target tens of thousands of dollars in cloud savings by reducing idle ECS capacity, improving scaling discipline, and preserving workload reliability through task-protection and monitoring-driven controls.

Technologies: Java, AWS SDK, Amazon ECS, ECS Fargate, SQS, CloudWatch metrics and alarms, RDS/SQL, Docker containers, service auto scaling, ECS Task Protection API, IAM, CloudFormation/Terraform patterns.

06

CUSTOMER EXPERIENCE
TaxAct
Scheduling
Expert platform
Calendar integration

XPert Booking and Scheduling Platform

Tax expert appointment, calendar, and scheduling platform delivery

Role: Served as Principal Engineer, technical lead, solution architect, and product-technology advisor for the implemented XPert booking and scheduling platform.

Problem: TaxAct needed to move beyond MS Bookings limitations exposed in TY 2021, including appointment volume, data fidelity, account management, granular permissions, calendar consolidation, expert assignment, real-time analytics, and routing.

Approach: Led the technology direction from platform evaluation into implementation, aligning operational requirements, vendor/platform capabilities, calendar integration, appointments, APIs, analytics, reporting, administration, staffing, Outlook/Google/iCal sync, reminders, permissions, and rules-based scheduling.

Impact: Delivered a stronger expert scheduling capability that improved customer-to-expert appointment workflows, reduced manual coordination, strengthened capacity planning, and supported branded appointment experiences, communications, rescheduling, and expert allocation.

Technologies: Booking and scheduling platforms, MS Bookings, Calendly, SimplyBook.me, Zoho Bookings, 10to8, Acuity, Exchange/Outlook, Google Calendar, iCal, API evaluation, SSO, MFA, role-based access, analytics and reporting.

07

DIGITAL INNOVATION
Virtual assistant
Recruiting
AI
Automation

Virtual Assistant Recruiting and Employee Automation Platform

AI assistant delivery for recruiter productivity and employee engagement

Role: Served as Principal Engineer, technical lead, solution architect, and digital innovation strategist for the implemented virtual assistant platform.

Problem: Express needed to improve recruiter productivity, reduce repetitive administrative work, support employee engagement at scale, and modernize digital employee experiences in a competitive talent market.

Approach: Delivered a virtual-assistant solution using rules-based automation, machine learning, natural-language interaction, human-centered design, MVP delivery cycles, and chatbot implementation patterns to support reminders, task assistance, social-media inbox triage, knowledge access, recommendations, and workflow automation.

Impact: Augmented staff recruiters and employees with an always-available assistant that reduced repetitive work, improved throughput, scaled support interactions, and created a feedback loop for continuous bot delivery and employee experience improvement.

Technologies: Virtual assistants, conversational AI, intelligent automation, machine learning concepts, NLP, chatbot UX, human-centered design, MVP planning, workflow automation, knowledge management, analytics, employee experience.

08

DIGITAL INNOVATION
Chatbots
CHIP / AVA
Employee self-service
Conversational AI

AI Chatbots CHIP and AVA Talent Engagement Platform

Conversational AI delivery for employee support and recruiting

Role: Served as Principal Engineer, technical lead, solution architect, and innovation strategist for implemented employee-facing chatbot platforms supporting talent engagement.

Problem: Express needed a scalable way to maintain candidate and employee connection across a large placement base while supporting modern expectations for fast, conversational, always-on digital service.

Approach: Implemented CHIP and AVA chatbot capabilities connecting employee self-service, recruiting support, organizational content search, recommendations, skills and experience profiling, natural-language interfaces, SMS/web delivery channels, backend-system access, and indexed knowledge sources.

Impact: Improved chat-based employee support and recruiting automation, made organizational knowledge more accessible, expanded self-service interactions, and helped recruiters focus on higher-value relationship work.

Technologies: Conversational AI, chatbots, NLP, Microsoft Bot Framework concepts, Azure Cognitive Services concepts, SMS and web channels, search and recommendation engines, knowledge indexing, employee profiles, recruiting automation, workflow integration.

09

FEATURED

Applied AI
Agentic workflow
Evaluation
Human review

RunbookOps AI | Reviewable incident-triage workflow

Personal proof-of-concept repository

Role: Served as Principal Engineer, technical lead, designer, and builder for a public GitHub proof of concept demonstrating hands-on AI engineering capability.

Problem: Incident reports can be noisy and hard to convert into consistent, reviewable triage recommendations.

Approach: Built a Python/Streamlit prototype using approved runbook retrieval, OpenAI Responses API, Pydantic validation, deterministic severity calibration, human-review routing, evaluation checks, reviewer feedback capture, and Ruff/pytest automation.

Impact: Demonstrates practical use of retrieval, structured model output, validation, evaluation, human oversight, and maintainable AI delivery practices.

Technologies: Python, Streamlit, OpenAI Responses API, Pydantic, pytest, Ruff.

Note: Prototype only. No customer production adoption is claimed.

10

FEATURED

Applied AI
Model routing
FinOps
API governance

CostOps Gateway and LLM Evaluation | Model execution controls

Personal proof-of-concept repository

Role: Served as Principal Engineer, technical lead, designer, and builder for a public GitHub proof of concept for cost-aware LLM execution controls.

Problem: Teams adopting LLMs need model routing, budget checks, reliable contracts, usage accounting, and testable provider behavior before AI workflows can scale responsibly.

Approach: Built a FastAPI gateway prototype with configurable routing, pre-call budget checks, bounded retries, Pydantic contracts, mock/OpenAI adapters, API-key protection, SQLite usage accounting, Streamlit reporting, and tests for routing, budgets, retries, ledger behavior, and reporting.

Impact: Shows the governance and platform thinking needed to move model calls from experiments toward controlled application infrastructure.

Technologies: FastAPI, Pydantic, SQLite, Streamlit, OpenAI adapter patterns, pytest.

Note: Prototype only. No customer production adoption is claimed.

11

FEATURED

Engineering management
AWS
Serverless
Customer experience
Data import

TaxAct XPert Assist, XPert Full-Service, and Data Import Platforms

TaxAct | Consumer tax software platforms

Role: Served as Principal Engineer and technical lead for architecture and delivery across flagship customer-facing tax, expert-assist, and data-import experiences.

Problem: Customers needed faster, more accurate ways to complete tax returns, reduce manual entry, and connect with tax experts during the filing journey.

Approach: Led digital transformation roadmaps, coached teams through delivery challenges, and advanced decoupled UI slices, containerization, distributed architectures, AWS serverless patterns, stateless event-driven services, and CQRS-oriented data segregation.

Impact: Improved platform agility, supported more accurate data import, reduced manual entry, and helped teams modernize customer-facing tax workflows while maintaining delivery discipline.

Technologies: AWS Lambda, Amazon RDS, S3, CloudWatch, Azure DevOps, Confluence, CQRS patterns, containers, event-driven services.

12

FEATURED

Machine learning
Azure
Operational analytics
Logistics optimization

AI/ML Well Phase Prediction and Logistics Analytics Platform

Global oil and gas program

Role: Served as Principal Engineer, technical lead, and solution architect for a predictive analytics platform within a large drilling, completions, and logistics program.

Problem: The client needed to combine operational data sources to forecast well phase completion, plan required field activities, improve logistics decisions, and monitor rig activity in near real time.

Approach: Designed Azure-based data and application patterns across .NET Core, Angular, Azure Functions, Data Factory, Databricks, Data Lake, SQL Server, Azure DevOps, and predictive analytics workflows.

Impact: Enabled better planning for rig schedules, job details, logistics demand, freight cost control, supplier operations, and operational productivity measures.

Technologies: .NET Core, Angular, Azure Functions, Azure Data Factory, Azure Databricks, Azure Data Lake, SQL Server, Azure DevOps, Power BI.

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FEATURED
Demand-to-Deliver
Azure
Logistics
Value chain

Demand Connect and Deliver Program

Digitally enabled value-chain and logistics platform

Role: Served as Principal Engineer, technical lead, and solution architect for the Demand Connect product vision, architecture, and delivery roadmap.

Problem: The program needed to break cross-functional silos across demand, order and procure, delivery, and usage feedback while giving teams one source of truth for materials, services, rig activity, and logistics planning.

Approach: Shaped the Demand Connect product vision, future-state ordering journey, Azure-based architecture, and feature roadmap across rig schedules, job details, logistical needs, supplier coordination, activity planning, Power BI integration, and business-partner workflows.

Impact: Supported a digitally enabled value chain intended to unlock more than $1B in value through integration, data quality, analytics-driven insights, automated workflows, and better logistics and supplier coordination.

Technologies: Azure API Management, Angular user experiences, Azure Functions, Service Fabric, Azure Data Factory, Azure Data Lake, DemandDB, WellView, PetroLink, Primavera, Power BI.

14

FEATURED
Cloud modernization
Scaled Agile
Angular
.NET Core
Azure

Cloud Modernization of Loan Origination Platform

Large mortgage technology platform

Role: Served as Principal Engineer, technical lead, solution architect, and senior program lead across a cloud modernization program involving many scrum teams.

Problem: The client needed to transition an aging loan origination platform to a cloud-based single-page application model while coordinating delivery across a large program.

Approach: Led bottom-up estimation, staffing analysis, PI planning, solution design, requirements analysis, future-state system design, deployment approach, and P1 escalation support. Helped define unit and integration testing strategies for source-code validation.

Impact: Supported transition to a modern cloud model, improved delivery coordination across 26 scrum teams, and helped establish practical architecture and release practices for a high-scale mortgage platform.

Technologies: Angular, .NET Core, Redis, SQL Azure, cloud-enabled architecture, SAFe/PI planning, unit and integration testing.

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FEATURED

Cloud data platform
Analytics
Power BI
Data lake

Business Intelligence Platform Modernization

Major oil-field services organization

Role: Served as Principal Engineer, technical lead, and solution architect from envisioning through solution design and implementation oversight.

Problem: The client needed a maintainable analytics platform that could improve operational reporting, reduce reporting turnaround time, and give leaders better insight into field operations.

Approach: Led architecture for a cloud-based data factory, data lake, analysis services cube, Power BI dashboards, and PowerApps feedback loops. Defined ingestion, transformation, storage, retrieval, security, performance, and knowledge-transfer patterns.

Impact: Delivered a more complete end-to-end analytics platform and helped move the client toward a maintainable reporting operating model.

Technologies: Azure Data Factory, Azure Data Lake, Analysis Services, Power BI, PowerApps, data modeling, ETL, security and performance patterns.

16

FEATURED

Healthcare billing
OCR
Workflow automation
SharePoint

eFiles Lockbox Processor and e-Vision OCR

Healthcare billing operations

Role: Served as Principal Engineer, technical lead, solution architect, and delivery lead for a healthcare billing automation solution.

Problem: Daily electronic lockbox processing required reliable handling of checks, remittance documents, invoices, explanation-of-benefits files, envelopes, and metadata.

Approach: Designed a solution with services for payload extraction, queue-based processing, OCR scanning, searchable PDF generation, metadata capture, and SharePoint storage.

Impact: Automated routing and retention of healthcare billing documents, reduced reliance on paper retention, and improved searchability and downstream billing operations.

Technologies: Custom .NET services, OCR, SharePoint, workflow queues, metadata capture, searchable PDF generation.

DIGITAL INNOVATION AND WORKFORCE PLATFORMS

Case studies are Word-native, editable, and ready to refine for GitHub publication

17

INNOVATION

Digital innovation
Roadmaps
MVP strategy
Executive workshops

Express Digital Innovation Strategy and Application Modernization

Large staffing and employment enterprise

Role: Served as technical lead, solution architect, digital innovation strategist, and delivery lead.

Problem: The client needed to respond to digital disruption, improve talent sourcing, modernize operations, and define a practical roadmap for new digital products and operating models.

Approach: Facilitated leadership workshops, ran envisioning cycles, defined operating-model and architecture roadmaps, shaped solution blueprints, staffed delivery teams, and presented prototypes and solution designs to senior leadership.

Impact: Created or shaped roughly 35 innovation opportunities, including AI-powered recruiting, employee career-path analytics, digital upskilling, automation, and process optimization concepts.

Technologies: Design thinking, innovation workshops, enterprise architecture, Agile delivery, MVP planning, analytics, automation, workforce platforms.

18

INNOVATION

Payroll
Timekeeping
SMS workflow
Process automation

SMS Time Reporting and Payroll Workflow Modernization

Staffing workforce operations

Role: Served as technical lead for the solution strategy and modern timecard concept for reducing paper-based time reporting.

Problem: Paper timecards created cost, delay, manual re-keying, payroll and invoice corrections, associate follow-up, and operational drag across recurring pay cycles.

Approach: Shaped a PIN-based SMS timecard concept with automated clock-ins, clock-outs, timecard verification, submission, reminders, opt-in and opt-out flows, and role-based reporting.

Impact: Outlined a path from paper forms toward more reliable, scalable, accurate time submission and payroll readiness.

Technologies: SMS workflows, reminders, PIN authentication, Q4 integration concepts, Power BI reporting, campaign automation concepts.

19

INNOVATION

Digital forms
Master data
Mobile workflows
Secure storage

E-Documents Initiative and Master Data Foundation

Staffing onboarding and associate operations

Role: Served as technical lead for the pilot and phased delivery model for digital employee forms and associate paperwork.

Problem: Associates and recruiters needed a faster, more secure, more centralized way to complete, sign, review, approve, and correct onboarding paperwork.

Approach: Shaped the e-documents MVP around mobile W4 and I9 form completion, mobile signatures, support-document capture, recruiter review and approval, encrypted storage, forms APIs, expiration logic, and a longer-term master data foundation for a single person record.

Impact: Defined a 12-to-15 week pilot path that could reduce duplicate work, improve data fidelity, centralize audit and control, and help place associates with customers faster.

Technologies: Mobile app features, web review portal, secure storage, APIs, authentication and authorization, master data services, ETL, validation rules.

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INNOVATION

Conversational AI
Digital workplace
RPA
Knowledge systems

Enterprise Virtual Assistant Concept

Staffing digital workplace and recruiter productivity

Role: Served as technical lead for the virtual assistant concept and pilot approach for supporting recruiters and employees.

Problem: Recruiters needed help managing everyday tasks such as unfilled jobs, candidate matches, sales calls, reminders, inbox messages, and operational follow-up.

Approach: Defined a virtual assistant model that could answer knowledge questions, recommend next actions, learn business rules, integrate with back-office systems, and escalate complex cases to human experts.

Impact: Created a phased approach for workshops, feasibility assessment, architectural blueprint, pilot development, bot training, roadmap development, and business-case validation.

Technologies: Natural language processing, knowledge base, semantic model, bots, RPA concepts, enterprise systems integration, asynchronous messaging.

Note: Source materials present this as strategy and pilot planning. Production implementation details should be validated before publication.

21

INNOVATION

Adoption analytics
Employee engagement
O365
Business value

Digital Workplace Insights

Modern workplace analytics and value realization

Role: Served as technical lead for the workplace analytics story connecting collaboration adoption, digital behaviors, and measurable business value.

Problem: Organizations were investing in workplace technology without measuring adoption, behavior change, or business outcomes at the level needed to guide action.

Approach: Framed metrics across adoption, behavior, value realization, employee engagement, collaboration, communication, search, file sharing, and workforce-specific business outcomes.

Impact: Presented a model for turning workplace data into targeted change efforts, improved adoption, and better return from digital workplace investments.

Technologies: Office 365, Teams, OneDrive, SharePoint, Yammer, Skype, digital workplace telemetry, KPI frameworks, adoption analytics.

ADDITIONAL ENTERPRISE MODERNIZATION WORK

Case studies are Word-native, editable, and ready to refine for GitHub publication

22

MODERNIZATION

O365 migration
Hybrid search
SharePoint
Disaster recovery

Enterprise Content and Collaboration Modernization

Energy enterprise collaboration platform

Role: Served as Principal Engineer, technical lead, solution architect, and delivery lead.

Problem: The client needed to modernize an aging SharePoint collaboration and application platform while preserving large volumes of content and business-critical custom applications.

Approach: Led delivery across multiple workstreams, designed SharePoint farms, disaster recovery patterns, hybrid enterprise search, custom .NET integrations, PeopleSoft profile synchronization, reporting, self-service BI components, responsive design, and custom branding.

Impact: Supported migration of 5.5TB of content, modernization of collaboration capabilities, restructuring of enterprise knowledge assets, and re-architecture of 200+ custom web applications across more than 30 operating units.

Technologies: SharePoint 2010/2016, Office 365, BA Insight, .NET integrations, hybrid search, PeopleSoft integration, disaster recovery.

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MODERNIZATION

Healthcare
ECM
Disaster resiliency
Public web modernization

Medicaid and Medicare Program Platform Modernization

State-funded healthcare program administrator

Role: Served as Principal Engineer, technical lead, solution architect, digital innovation strategist, and delivery lead.

Problem: The client needed to modernize a legacy public platform, improve user experience, increase resiliency, and support healthcare program delivery requirements.

Approach: Helped lead migration of legacy content, development of a new search center, re-architecture of custom web applications and web parts, and design of a disaster recovery solution.

Impact: Improved online presence, functionality, user experience, and continuity of service for a healthcare program administrator.

Technologies: SharePoint, search, disaster recovery, UX, application development, BI and analytics, Agile delivery.

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MODERNIZATION

Salesforce integration
Document workflow
SharePoint
Automation

Salesforce Document Management and Workflow Platform

Global telecom sales-cycle platform

Role: Served as Principal Engineer, technical lead, solution architect, and delivery lead.

Problem: The sales organization needed automated document repositories, workflows, metadata, permissions, and quote/contract artifacts connected to Salesforce entities.

Approach: Designed automated near-real-time provisioning for Account, Opportunity, Offer, and Contracted Quote repositories with predefined document libraries, metadata correlation, workflows, and rules-based access.

Impact: Created a document and collaboration platform for 50,000+ site collections, improved document traceability, reduced manual entry, and gave sales teams better search and visibility into transaction artifacts.

Technologies: Salesforce, SharePoint, metadata, workflows, rules-based permissions, document management, search.

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MODERNIZATION

Enterprise search
Metadata
Accessibility
Usage analytics

Enterprise Search Modernization

Global consumer goods enterprise

Role: Served as Principal Engineer, technical lead, solution architect, and delivery lead.

Problem: The organization needed a scalable universal search experience across collaboration, line-of-business, partner, and public data sources.

Approach: Designed, customized, and deployed enterprise search using metadata, content types, managed properties, refiners, query rules, result sources, accessibility practices, and usage/performance analytics.

Impact: Supported crawl indexes exceeding 40 million documents and sites and improved access to business-critical knowledge assets.

Technologies: SharePoint Enterprise Search, metadata, managed properties, refiners, query rules, usage analytics, accessibility.

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MODERNIZATION

Workflow
Batch processing
Business rules
Financial data

Enterprise Workflow and Batch Framework

Investment management operations platform

Role: Served as Principal Engineer, technical lead, solution architect, and delivery lead.

Problem: The client needed to modernize a proprietary accounting and results system used by operations and investment groups.

Approach: Designed a configurable business-rules and batch-processing platform for global allocations, positions, real-time financial calculations, account/fund-level transactions, aggregate views, integrations, and complex data volumes.

Impact: Helped streamline manual processes, improve insight into financial data, and support one of the client's most ambitious custom solution efforts.

Technologies: Custom application architecture, business rules, batch processing, financial data integration, scalable data design.

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MODERNIZATION

AI history
Knowledge acquisition
Ontologies
Expert systems

Athena Knowledge Management and Learning System

AI and cognitive computing platform

Role: Served as Principal Engineer, technical lead, solution architect, and lead programmer.

Problem: The platform needed a way for subject-matter experts to structure knowledge that could train a proprietary AI processor.

Approach: Architected a rich internet knowledge-acquisition and learning platform with taxonomy, ontology, user-experience, automated testing, and continuous-integration considerations.

Impact: Created an early hands-on AI-oriented delivery story that connects expert knowledge, structured learning data, and application engineering.

Technologies: Rich internet application architecture, taxonomies, ontologies, expert knowledge capture, automated testing, CI.

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MODERNIZATION

Vendor management
Risk
Contract oversight
Reporting

VendorView Vendor and Contract Management Platform

Banking acquisition integration

Role: Served as Principal Engineer, technical lead, and solution architect.

Problem: The organization needed a better way to track vendors, contracts, service relationships, RFP responses, risk reviews, and executive reporting during acquisition integration.

Approach: Designed a web-based vendor and contract management application with vendor selection, RFP response tracking, risk assessment, review checklists, reporting, and role-based access.

Impact: Improved visibility into vendor and contract data, supported acquisition rationalization, and reduced operational friction around vendor oversight.

Technologies: Web application architecture, role-based access, reporting, vendor data, contract workflows.

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MODERNIZATION
Healthcare
Cloud migration
Business process
Reporting

HMO Cloud Migration and Business Process Modernization

Healthcare operations transformation

Role: Served as Principal Engineer, technical lead, solution architect, and delivery lead.

Problem: A Medicare Advantage organization serving more than 165,000 members needed to migrate legacy systems toward cloud-based business process and automation capabilities.

Approach: Supported migration planning, report conversion, structural and data-mapping changes, team coordination, client feedback management, and delivery cadence.

Impact: Helped move mission-critical reporting and operational workflows toward a more scalable, cloud-oriented business process platform.

Technologies: Cloud migration planning, enterprise reporting, data mapping, Agile delivery, business process automation.

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MODERNIZATION
App catalog
SharePoint
Personalization
Enterprise search

Veritas Internal App Store

Enterprise application discovery platform

Role: Served as Principal Engineer, technical lead, solution architect, and delivery lead.

Problem: Employees needed an easier way to discover, access, review, and share hundreds of internal applications organized by business purpose and function.

Approach: Designed a SharePoint-hosted app store with browsing, favorites, app collections, access requests, launch paths, recommendations, featured apps, reviews, and business-oriented categorization.

Impact: Improved internal application discoverability, reduced friction in finding relevant tools, and supported knowledge sharing around enterprise applications.

Technologies: SharePoint, app catalog design, metadata, personalization concepts, enterprise search, user experience design.

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MODERNIZATION
CMS migration
Intranet
UX design
Disaster recovery

Enterprise CMS and Intranet Portal Modernization

Corporate intranet migration and employee self-service

Role: Served as Principal Engineer, technical lead, solution architect, migration lead, UX designer, and delivery lead.

Problem: The client needed to migrate an aging CMS and intranet platform while supporting personal sites, corporate communications, collaboration sites, and employee self-service workflows.

Approach: Led design, migration, UX, custom branding, search-center development, application modernization, and resiliency planning for a modern SharePoint implementation.

Impact: Supported migration of more than 15,000 user accounts and personal sites, improved intranet usability, and modernized collaboration and self-service capabilities.

Technologies: SharePoint, enterprise CMS, intranet services, custom branding, search, disaster recovery, UX design.

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BI AND KNOWLEDGE
Self-service BI
Data integration
Security
Enterprise analytics

Self-Service Business Intelligence Enterprise Platform

Internal BI architecture and secure data collaboration

Role: Served as Principal Engineer, technical lead, solution architect, and delivery lead.

Problem: A global enterprise needed a unified BI solution architecture that could enable information sharing across operating platforms in a secure, collaborative environment.

Approach: Led design and implementation of a scalable BI platform, working with stakeholders to define requirements and shape data integration, profiling, cleansing, transformation, access control, and audit patterns.

Impact: Helped establish a self-service BI capability that accelerated data flow, improved collaboration, and supported better decision-making across business units.

Technologies: Business intelligence architecture, data integration, data quality, security controls, access management, analytics platform design.

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BI AND KNOWLEDGE
BI portal
Report inventory
Role-based access
Decision support

Consolidated Business Intelligence and Report Distribution Platform

Cross-functional information sharing portal

Role: Served as Principal Engineer, technical lead, solution architect, and delivery lead.

Problem: Business users needed a single way to inventory, locate, secure, and share reports, scorecards, dashboards, and data lists without consolidating every reporting system or data repository.

Approach: Designed and led delivery of a cross-functional information-sharing portal with role-based permissions, report inventory patterns, and guidance for development teams.

Impact: Improved access to business data assets, reduced time spent locating reports, strengthened information governance, and supported faster leadership decision-making.

Technologies: BI portal architecture, report distribution, role-based security, metadata, collaboration, enterprise reporting.

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BI AND KNOWLEDGE
Travel analytics
Compliance
Cost forecasting
CO2 reporting

Travel BI Reporting and Analytics Solution

Travel spend, compliance, and sustainability analytics

Role: Served as Principal Engineer, technical lead, solution architect, and delivery lead.

Problem: Leadership needed a reporting solution to monitor, assess, and forecast more than $50 million in annual travel spend while also tracking regional policy compliance and carbon footprint indicators.

Approach: Designed a reporting and analytics application with metrics for missed savings, booking dates, advanced ticketing, upgrades, compliance, cancellations, and sustainability-related measures.

Impact: Delivered a scalable analytics solution that helped monitor travel spend, improve policy visibility, and support cost and sustainability management.

Technologies: SQL Server, SSRS, SSIS, MDX, reporting dashboards, data transformation, SharePoint project documentation.

GITHUB PUBLICATION BACKLOG

Suggested next steps for turning this into a living portfolio